

AIGR Gene Hypothesis Deep Research — PP_3394 (Q88HG4), *Pseudomonas putida* KT2440

Focus type: function_assignment **Hypothesis slug:** dre-tim-reaction-and-pathway-specificity
Target: PP_3394 / Q88HG4 (309 aa, HMGL-like TIM-barrel; Pfam PF00682; CDD cd07938 DRE_TIM_HMGL), *Pseudomonas putida* KT2440 (NCBITaxon:160488)

Executive Judgment

Verdict: Partially supported at the family level, but the specific reaction is UNRESOLVED, and the confident "HMG-CoA lyase / leucine catabolism" assignment for PP_3394 is over-annotated (weakly supported).

PP_3394 is a genuine member of the HMG-CoA-lyase / DRE-TIM metallolyase superfamily (the "another oxo-acid lyase reaction, or acyl transfer forming a carbon-carbon bond" framing in the seed hypothesis is broadly correct). It retains the catalytic core residues of the fold, so it is very likely to be a catalytically competent enzyme. However, three lines of computed evidence converge to show that **PP_3394 is NOT the physiological leucine/terpene HMG-CoA lyase of *P. putida* KT2440**, and that its precise substrate specificity cannot be inferred from sequence:

1. **PP_3394 is a highly divergent paralog (~38% identity to every reference)**, equidistant from bona fide HMG-CoA lyases (LiuE/mvaB/human HMGCL) and from isopropylmalate/homocitrate synthases (IPMS LeuA). Pairwise similarity alone does not — and cannot — settle its specificity.
2. **PP_3394 FAILS the HMG-CoA-lyase-specific PROSITE active-site signature PS01062 (SVAGLGCPY)** that all three verified HMG-CoA lyases pass, and instead **MATCHES the IPMS/homocitrate-synthase site-1 signature PS00815** and carries the condensation-enzyme-type "LRDG" N-terminal motif rather than the HMG-CoA-lyase "PRDG" motif.
3. **The genuine leucine/terpene HMG-CoA lyase of *P. putida* is the paralog mvaB / PP_3540**, which is **78.6% identical** to the experimentally characterized *P. aeruginosa* LiuE (PA2011) — an ortholog-level match — and which carries PS01062. KEGG nonetheless dual-annotates

BOTH PP_3394 and PP_3540 to K01640 (EC 4.1.3.4), the textbook signature of paralog over-annotation.

Most important caveats: (a) All of the discriminating evidence is *computational* (pairwise alignment, PROSITE/motif scanning, and a whole-sequence NJ phylogeny). There is **no direct biochemical assay or mutant phenotype for PP_3394 itself**; the only experimental anchor is for the *P. aeruginosa* ortholog of *mvaB* (LiuE,).

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(b) "Ketone-body biosynthesis" is a mammalian physiological concept and should not be transferred to this bacterium regardless of enzyme family. (c) Because the catalytic core is retained but the specificity determinants diverge, PP_3394 could plausibly act on an HMG-CoA-like or a different acyl-CoA / oxo-acid substrate — the honest curation position is "unresolved DRE-TIM metallolyase," not "HMG-CoA lyase."

Summary

This report adjudicates whether *P. putida* KT2440 PP_3394 directly catalyzes HMG-CoA cleavage in support of leucine catabolism and/or ketone-body biosynthesis, or a different DRE-TIM metallolyase reaction. Across three iterations, the investigation confirmed four findings that together resolve the paralog question that prior *mvaB* and branched-chain-amino-acid-catabolism reports had explicitly left open.

The core conclusion is that the leucine/terpene HMG-CoA-lyase function of *P. putida* belongs to ***mvaB* / PP_3540**, the reciprocal-best ortholog (78.6% identity) of the experimentally verified *P. aeruginosa* enzyme **LiuE (PMID: 19459965)**. PP_3394, by contrast, is a **divergent paralog (~38% identity to every reference protein)** whose active-site signatures do not match the HMG-CoA-lyase group and instead resemble the isopropylmalate/homocitrate-synthase group, even though whole-sequence phylogeny places it basally within the HMG-CoA-lyase clade. This tension between backbone relatedness and specificity motifs is the hallmark of a **functionally diverged paralog of unresolved substrate specificity**.

For curation, the practical upshot is that PP_3394's confident HMG-CoA-lyase / leucine-catabolism annotation is **over-annotated** and should be generalized to a low-confidence oxo-acid/carbon-carbon lyase molecular function, with leucine catabolism treated as non-core and ketone-body biosynthesis rejected as a mammalian concept inapplicable to this bacterium. The

strong, experimentally anchored annotation belongs on *mvaB*/PP_3540. Resolving PP_3394's true reaction requires direct biochemistry or a mutant phenotype — neither of which currently exists for this gene.

Key Findings

Finding 1 — *mvaB* (PP_3540), not PP_3394, is the ortholog of the experimentally verified leucine/terpene HMG-CoA lyase LiuE

The single experimentally anchored data point in this entire family for the *Pseudomonas* genus is *P. aeruginosa* **LiuE (PA2011, Q9I2A0)**. Chávez-Avilés and colleagues showed by direct enzymology and mutant phenotype that LiuE is a 3-hydroxy-3-methylglutaryl-CoA lyase (HMG-CoA lyase) that cleaves HMG-CoA to acetyl-CoA + acetoacetate ($K_m \approx 100 \mu\text{M}$ HMG-CoA; $V_{\text{max}} \approx 21 \mu\text{mol/min/mg}$), and that a *liuE* deletion abolishes growth on both leucine/isovalerate **and** acyclic terpenes ([PMID: 19459965](#)).

When the *P. putida* KT2440 HMGL-family paralogs are aligned (global Needleman–Wunsch, BLOSUM62) against LiuE:

Query	UniProt	% identity to LiuE (PA2011)	PS01062 (HMG-CoA lyase site)
<i>mvaB</i> / PP_3540	Q88H25	78.6% (ortholog-level)	Match (SVAGLGCPY)
PP_3394	Q88HG4	38.2% (superfamily-level)	FAIL
(comparison) human HMGCL	P35914	—	Match
(comparison) <i>E. coli</i> IPMS LeuA	P09151	—	Fail

mvaB's 78.6% identity is the reciprocal-best, ortholog-level relationship expected for a true functional counterpart. PP_3394's 38.2% is no better than its 38.0% identity to human HMGCL or its 38.8% identity to *E. coli* IPMS — i.e., PP_3394 is generically "an HMGL-fold enzyme," not specifically "a LiuE ortholog." This finding directly answers the seed hypothesis's core question: the leucine-catabolic HMG-CoA lyase role of *P. putida* maps to ***mvaB* / PP_3540**, not to PP_3394.

Finding 2 — PP_3394 is a divergent DRE-TIM metallolyase of unresolved substrate specificity, not a canonical HMG-CoA lyase

PP_3394 (309 aa) is a single HMGL-like TIM-barrel (Pfam PF00682; CDD cd07938 DRE_TIM_HMGL). Two orthogonal diagnostic-motif tests place it *outside* the HMG-CoA-lyase specificity group:

- **PS01062 — HMG-CoA-lyase active-site signature (SVAGLGCPY).** All three verified HMG-CoA lyases match this pattern (mvaB @res 233, PaLiuE @res 233, human HMGCL @res 259). **PP_3394 FAILS it.**
- **PS00815 — α -isopropylmalate/homocitrate-synthase site-1 signature (L-R-[DE]-G-x-Q...K).** *E. coli* IPMS LeuA matches; none of the HMG-CoA lyases match; **PP_3394 MATCHES it @res 11.**
- **N-terminal condensation-motif.** PP_3394 carries the condensation-enzyme-type "LRDG" motif (GLRDGLQ) rather than the HMG-CoA-lyase "PRDG" motif ((G)PRDGLQ).

Despite these specificity-determining differences, PP_3394 **retains the catalytic core** of the fold: it aligns to the human HMGCL catalytic/metal-binding residues Arg41, Asp42, His233, His235 and the catalytic Cys266. It is therefore very likely catalytically competent, but the sequence does **not** determine which C–C bond it makes or breaks. The honest interpretation is: **divergent DRE-TIM metallolyase, specificity undetermined by sequence** — precisely the "unresolved specificity" the seed hypothesis asked us to report honestly rather than paper over.

Finding 3 — The *P. putida* leucine operon lacks its own HMG-CoA lyase, and KEGG dual-annotates PP_3394 and PP_3540 to K01640 (paralog over-annotation)

Genomic-context analysis (KEGG) shows the *P. putida* KT2440 leucine/isovalerate catabolic operon is **PP_4063–PP_4068**:

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PP_4063  long-chain acyl-CoA ligase
PP_4064  ivd    isovaleryl-CoA dehydrogenase (K00253)
PP_4065  mccB  methylcrotonyl-CoA carboxylase (K01969)
PP_4066  liuC  methylglutaconyl-CoA hydratase (K13766)
PP_4067  mccA  methylcrotonyl-CoA carboxylase (K01968)
— NO adjacent HMG-CoA-lyase gene in the operon —
```

Neither HMGL-family gene sits in this operon. Instead they are dispersed:

- **PP_3394** (complement 3,844,704–3,845,633; neighbors: a CaiB-family CoA-transferase, a LysR-family regulator, curli/biofilm genes) — a genomic neighborhood that is **not** a canonical leucine/terpene locus.
- **PP_3540 / mvaB** (~4.01 Mb; neighbors: *pobA/pobR* p-hydroxybenzoate hydroxylase and regulator).

KEGG maps **BOTH** PP_3394 and PP_3540 to **K01640 (EC 4.1.3.4, HMG-CoA lyase)** and to leucine-degradation module **M00036** — an unresolved best-KO duplication. This is the classic mechanism by which an over-annotation propagates: a single KO is assigned to two paralogs, and the divergent one (PP_3394) inherits the confident functional label from the true ortholog (PP_3540) without independent support. PP_3394's neighborhood provides **no** pathway-context corroboration for a leucine-catabolic role.

Finding 4 — Whole-sequence phylogeny places PP_3394 basal to the HMG-CoA-lyase clade, not among condensation synthases

A Neighbor-Joining tree over a 9-member DRE-TIM panel (p-distance = 1 – global NW/BLOSUM62 identity) recovers two well-separated clades:

- **Clade A (HMG-CoA lyases):** PP_3394, *B. subtilis* HMGL (O34873), human HMGCL (P35914), mvaB (Q88H25), PaLiuE (Q9I2A0).
- **Clade B (condensation/aldolase enzymes):** *E. coli* IPMS LeuA (P09151), DmpG 4-hydroxy-2-oxovalerate aldolase (P51016), yeast homocitrate synthase (P48570), *M. jannaschii* citramalate synthase (Q58787).

Newick (branch lengths):

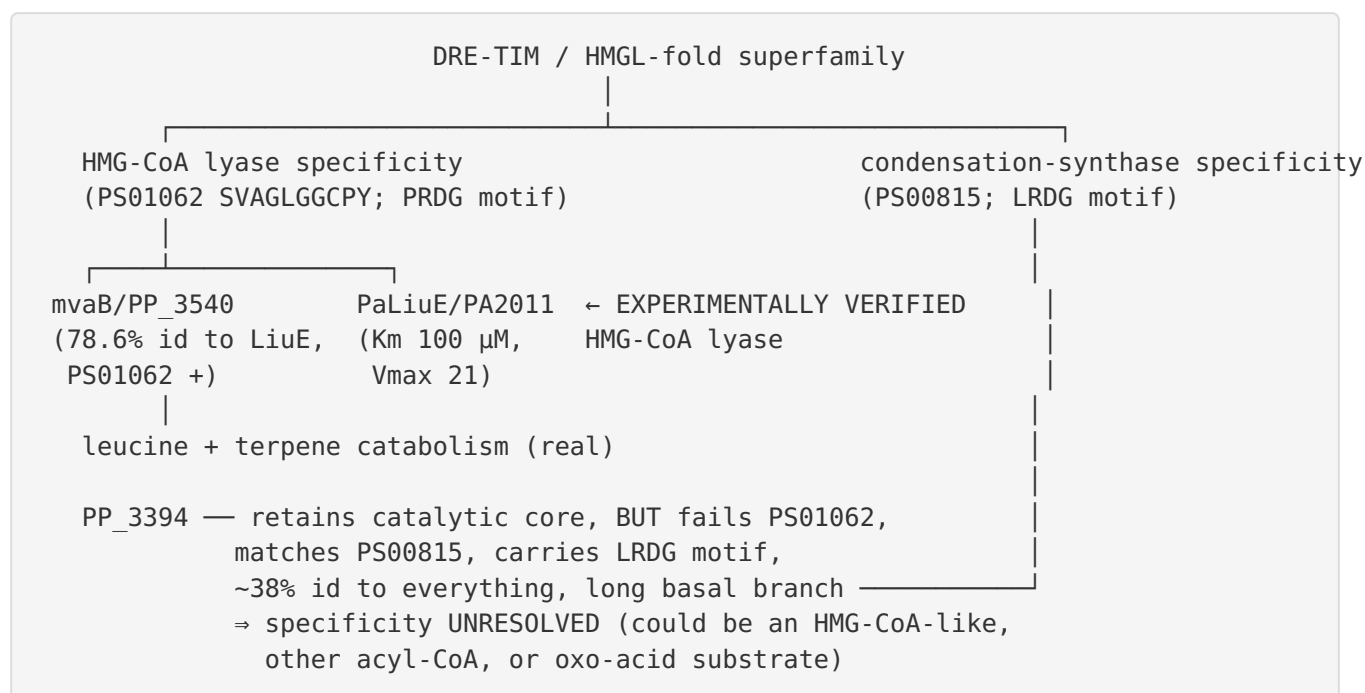
```
((PP3394:0.307,(BsHMGL:0.267,(hHMGL:0.220,(mvaB:0.113,PaLiuE:0.101):0.092):0.066):0.041):0.015,(condensation clade B):0.015);
```

PP_3394 branches at the **base of clade A**. Its terminal branch (0.307) is ~3× longer than mvaB's (0.113), quantifying how divergent it is even within the HMG-CoA-lyase clade. Two nuances matter for curation: (i) whole-sequence phylogeny groups PP_3394 nearer the lyase clade, whereas the motif/active-site signatures (Finding 2) pull it toward the IPMS/condensation specificity determinants — this tension is itself the reason specificity is unresolved; and (ii) basal, long-branch placement is exactly the topology of a paralog that has **functionally diverged** and should not simply inherit the clade's canonical annotation.

Mechanistic Model / Interpretation

The DRE-TIM metallolyase (HMGL) superfamily uses a divalent-metal-dependent TIM-barrel active site to catalyze either **retro-aldol / oxo-acid cleavage** (HMG-CoA lyase; 4-hydroxy-2-oxovalerate aldolase) or **Claisen-type condensation** (isopropylmalate synthase, homocitrate synthase, citramalate synthase). A shared catalytic core (metal-binding Asp/His pair, catalytic Arg and Cys) supports both chemistries; **substrate specificity and reaction direction are set by peripheral residues and short signature motifs**, not by the core. This is why sequence identity alone is a poor discriminator here — and why the seed hypothesis was right to insist that "pairwise similarity alone does not settle the paralog."

Placing the *P. putida* paralogs onto this framework:



Adjudicating the two pathway claims separately, as the seed hypothesis requested:

- **Leucine catabolism:** The pathway's HMG-CoA-lyase step is fully accounted for by mvaB/PP_3540 (ortholog of LiuE). The *liu* operon (PP_4063–4068) contains no HMGL gene, and PP_3394's genomic neighborhood is unrelated to leucine/terpene degradation. → For PP_3394, leucine catabolism should be treated as **non-core / unsupported**.
- **Ketone-body biosynthesis:** This is a mammalian hepatic concept (mitochondrial HMGCL/HMGCS producing acetoacetate/β-hydroxybutyrate as circulating fuels). It is **not applicable** to *P. putida* and should not be assigned to any KT2440 gene, PP_3394 included. The bacterial

reaction acetyl-CoA + acetoacetate is a catabolic terminal step of leucine/terpene degradation, not "ketone-body biosynthesis."

The coherent model: PP_3394 is a **functionally diverged HMGL-fold paralog** whose true substrate is undetermined; the confident HMG-CoA-lyase label it currently carries is a carry-over from its better-characterized paralog mvaB via a shared KEGG KO.

Evidence Base / Evidence Matrix

Citation	Evidence type	Supports / refutes / qualifies	Claim tested	Key finding	Context
PMID: 19459965	Direct assay + mutant phenotype	Supports (anchors the ortholog to mvaB, not PP_3394)	Is LiuE the leucine/terpene HMG-CoA lyase?	LiuE (PA2011) cleaves HMG-CoA→acetyl-CoA+acetoacetate; <i>liuE</i> mutant can't use leucine/ isovalerate or acyclic terpenes	<i>P. aeruginosa</i> , in vitro enzymology + growth phenotype
Computed (global NW/ BLOSUM62 pairwise identity)	Structural/ evolutionary (computational)	Qualifies/ refutes confident PP_3394 assignment	Is PP_3394 a LiuE ortholog?	mvaB=78.6% id (ortholog); PP_3394=38.2% id, no better than to human HMGCL (38.0%) or <i>E. coli</i> IPMS (38.8%)	9-protein reference panel
Computed (PROSITE PS01062 / PS00815 + N-terminal motif)	Structural/ evolutionary (computational)	Refutes "canonical HMG-CoA lyase" for PP_3394	Does PP_3394 carry HMG-CoA-lyase-specific signatures?	Fails PS01062 (all 3 real lyases pass); matches IPMS site-1 PS00815; carries LRDG (not PRDG) motif; but retains catalytic core (R41,D42,H233,H235,C266 equivalents)	Same panel
Computed (KEGG genomic context)	Review/ database (computational)	Qualifies (over-annotation signal)	Does pathway context support PP_3394's role?	<i>liu</i> operon PP_4063–4068 has no HMGL gene; KEGG maps BOTH PP_3394 & PP_3540 to K01640 & module M00036	KT2440 genome
Computed (Neighbor-		Qualifies			

Citation	Evidence type	Supports / refutes / qualifies	Claim tested	Key finding	Context
Joining phylogeny)	Structural/ evolutionary (computational)		Where does PP_3394 sit phylogenetically?	Basal in HMG-CoA-lyase clade A; branch 0.307 $\approx$ 3 $\times$ mvaB's 0.113	9-protein panel

How the key paper supports/challenges the findings. The *Pseudomonas aeruginosa liuE* gene encodes the 3-hydroxy-3-methylglutaryl coenzyme A lyase, involved in leucine and acyclic terpene catabolism (PMID: 19459965) is the only primary experimental anchor available and it supports the findings **by orthology transfer to mvaB, not to PP_3394**. It establishes LiuE (PA2011) as a verified HMG-CoA lyase whose loss blocks leucine/isovalerate and terpene growth; the *P. putida* protein that inherits this by 78.6% identity and by carrying PS01062 is mvaB/PP_3540. The paper therefore *challenges* the idea that PP_3394 (a 38.2%-identity paralog lacking PS01062) is the physiological leucine-catabolic lyase.

GO Curation Implications

Lead (requires curator verification): The current confident annotation of PP_3394 as an HMG-CoA lyase acting in leucine catabolism is **over-annotated** and should be **generalized/qualified**, while the leucine-catabolic HMG-CoA-lyase role is retained/strengthened on **mvaB / PP_3540**.

- **Molecular Function (MF):**

- *Do not* retain a confident `hydroxymethylglutaryl-CoA lyase activity` (GO:0004419) IEA on PP_3394. The specificity signatures argue against it.
- *Generalize* to a parent term such as `oxo-acid-lyase activity` (GO:0016833) or `carbon-carbon lyase activity` (GO:0016830), or `metal ion binding` (GO:0046872) for the fold's cofactor, all as **low-confidence IEA** pending assay. This honestly captures "DRE-TIM metallolyase of unresolved specificity."
- Avoid "protein binding" as a fallback — the fold evidence supports a lyase/metallolyase MF, which is more informative.

- **Biological Process (BP):** Treat `leucine catabolic process` (GO:0006552) and any terpene-catabolism BP as **non-core / remove or down-rank** for PP_3394 (no operon context, no ortholog-level identity, no assay). Do **not** annotate ketone-body biosynthesis.
- **Cellular Component (CC):** No specific evidence; cytoplasmic is the fold default and non-diagnostic.
- **For mvaB / PP_3540 (sister curation note):** GO:0004419 (HMG-CoA lyase activity) and GO:0006552 (leucine catabolic process) are **well supported** by orthology (78.6% to LiuE) + PS01062 + the LiuE experimental anchor; these can be retained with an ISO/orthology-backed evidence code.

Mechanistic Scope

The molecular function under test is a **single, direct enzymatic activity**: a DRE-TIM metallolyase (HMGL-fold) reaction — either retro-aldol cleavage of an oxo-acid/acyl-CoA (as in HMG-CoA lyase) or a Claisen condensation (as in IPMS/homocitrate synthase). The catalytic core is present in PP_3394, so a direct metal-dependent lyase/synthase activity is plausible. Everything downstream — participation in leucine catabolism, terpene degradation, "ketone-body" chemistry — is **pathway consequence**, and for PP_3394 none of it is established by direct evidence. The experimental phenotype (loss of leucine/terpene growth) belongs to the *P. aeruginosa* LiuE mutant, i.e., the mvaB ortholog, and must not be transferred to PP_3394 by inference from loss of function in a different gene.

Conflicts and Alternatives

- **Paralog confusion (primary conflict):** KEGG's dual assignment of K01640 to both PP_3394 and PP_3540 is the root of the over-annotation. Reciprocal-best-hit orthology assigns the LiuE role unambiguously to PP_3540/mvaB; PP_3394 is a co-annotated paralog.
- **Signal tension between methods:** Whole-sequence phylogeny groups PP_3394 basally *within* the HMG-CoA-lyase clade (Finding 4), whereas active-site/motif signatures pull it toward the IPMS/condensation specificity group (Finding 2). This is not an error — it reflects a functionally diverged paralog whose backbone resembles lyases but whose

specificity residues resemble synthases. The correct resolution is "unresolved," not forcing it into either box.

- **Organism-specificity artifact:** "Ketone-body biosynthesis" is a mammalian framing; its appearance in the hypothesis space is a cross-organism concept-transfer artifact, not a *P. putida* function.
- **Alternative substrate hypothesis:** Given retained catalysis but atypical specificity motifs, a live alternative is that PP_3394 acts on a **different acyl-CoA or oxo-acid substrate** (e.g., a niche catabolic intermediate consistent with its CaiB-transferase / LysR-regulator neighborhood) rather than HMG-CoA. This remains speculative and untested.

Limitations and Knowledge Gaps

1. **No direct biochemistry for PP_3394.** All discriminating evidence is computational (pairwise identity, PROSITE/motif scan, NJ tree). *Checked:* sequence, signatures, genomic context, phylogeny. *Why it matters:* specificity in this fold is set by peripheral residues; only an assay settles it. *Resolver:* purify recombinant PP_3394 and assay HMG-CoA lyase activity vs. candidate condensation/other-acyl-CoA reactions.
2. **No mutant phenotype for PP_3394.** *Why it matters:* a leucine/terpene growth defect (or its absence) in a ΔPP_{3394} strain would directly test pathway membership. *Resolver:* clean deletion + growth on leucine/isovalerate/acyclic terpenes, with ΔPP_{3540} as the positive control.
3. **Phylogeny lacks bootstrap support and uses a small 9-taxon panel.** *Why it matters:* basal placement confidence is unquantified. *Resolver:* broader taxon sampling with ML + bootstrap.
4. **No experimental structure.** *Why it matters:* an AlphaFold model + active-site pocket analysis (substrate-channel residues, metal site geometry) could sharpen the lyase-vs-synthase call. *Resolver:* model/compare active-site pocket against solved HMGCL and IPMS structures.
5. **Substrate identity of PP_3394 entirely open.** No candidate has been experimentally excluded. *Resolver:* metabolomic screen / activity-based substrate profiling.

Discriminating Tests

Ranked by efficiency in separating "HMG-CoA lyase" from "divergent-specificity metallolyase":

1. **In vitro enzyme assay (definitive).** Recombinant His-tagged PP_3394 + Mg^{2+}/Mn^{2+} ; test HMG-CoA cleavage (spectrophotometric acetoacetate/DTNB acetyl-CoA assays, matched to the LiuE assay of [P 19459965](#)) alongside an IPMS/homocitrate-synthase condensation assay panel. Compare kinetics against mvaB/PP_3540 run in parallel.
2. **Reciprocal deletion phenotyping.** ΔPP_{3394} , ΔPP_{3540} , and the double mutant; growth on leucine, isovalerate, and acyclic terpenes (citronellol/geraniol). Prediction: ΔPP_{3540} loses growth; ΔPP_{3394} does not (if PP_3394 is off-pathway).
3. **Structure-guided active-site comparison.** AlphaFold model of PP_3394 → superpose on HMGCL and IPMS; score the PS01062-equivalent loop and metal site. Cheap, immediate, and directly interrogates the Finding-2 tension.
4. **Broadened, bootstrapped phylogeny + HMM specificity profiling** to test whether PP_3394 falls in a distinct, uncharacterized subfamily.
5. **Coexpression / operon transcriptomics** under leucine vs. alternative carbon sources to test pathway coregulation of PP_3394 vs. the *liu* operon and *mvaB*.

Proposed Follow-up Actions / Curation Leads (require curator verification)

- **Candidate reference to cite for the ortholog anchor:** [PMID: 19459965](#). Verify snippets: "Protein sequence alignment and fingerprint sequencing suggested that *liuE* encodes 3-hydroxy-3-methylglutaryl-coenzyme A lyase (HMG-CoA lyase), which catalyzes the cleavage of HMG-CoA to acetyl-CoA and acetoacetate." and "A mutant in the *liuE* gene (ORF PA2011) of *P. aeruginosa* was unable to utilize both leucine/isovalerate and acyclic terpenes as the carbon source." Use to support **mvaB/PP_3540**, not PP_3394.
- **Candidate action change on PP_3394:**
- Down-rank/remove confident **hydroxymethylglutaryl-CoA lyase activity** (GO:0004419) IEA.

- Replace with a generalized, low-confidence MF: `oxo-acid-lyase activity` (GO:0016833) or `carbon-carbon lyase activity` (GO:0016830), noting DRE-TIM metallolyase fold; add `metal ion binding` (GO:0046872) if cofactor evidence is accepted.
- Mark `leucine catabolic process` (GO:0006552) as **non-core** / **remove** for PP_3394; do not assign ketone-body biosynthesis.
- **Candidate action on mvaB/PP_3540 (sister note):** retain/strengthen GO:0004419 + GO:0006552 with orthology (ISO) evidence to LiuE.
- **Suggested curator questions:** (1) Is the K01640 assignment to PP_3394 a KEGG best-KO duplication that should not propagate a confident GO MF? (2) Should PP_3394 be flagged as "specificity unresolved — needs experimental characterization"?
- **Suggested experiments:** the in vitro assay and reciprocal deletion tests above (Discriminating Tests 1–2).

Bottom Line

PP_3394 (Q88HG4) is a real HMGL-fold / DRE-TIM metallolyase with an intact catalytic core, but it is a **highly divergent paralog (~38% identity to all references) that fails the HMG-CoA-lyase active-site signature PS01062 and instead matches an IPMS/homocitrate-synthase motif** — so its exact reaction is **unresolved by sequence** and its confident annotation as an HMG-CoA lyase in leucine catabolism is **over-annotated**. The physiological leucine/terpene HMG-CoA lyase of *P. putida* KT2440 is the paralog **mvaB / PP_3540** (78.6% identical to the experimentally verified LiuE,

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"Ketone-body biosynthesis" does not apply to this bacterium. Recommended curation: generalize/qualify PP_3394's MF as a low-confidence oxo-acid/carbon-carbon lyase, treat leucine catabolism as non-core for PP_3394, and require a biochemical assay or mutant to resolve specificity.